Basbas, Junlie C. DATA STRUCTURE AND ALGORITHM

WMT 2-2 ACTIVITY 1

A. Discuss Briefly:

1. What is an algorithm, and how is it related to the tasks you perform in your daily life?

- An algorithm is a step-by-step set of instructions or procedures used to solve a problem or complete a specific task. In daily life, we use algorithms even without realizing it. For example, when preparing food, we follow steps such as gathering ingredients, cooking them, and serving the food. This shows that algorithms help us complete tasks in an organized and efficient way.

2. Why is it important for an IT student to learn algorithms?

- It is important for an IT student to learn algorithms because they are the foundation of programming and problem-solving. Algorithms teach students how to break a complex problem into smaller, manageable steps and find an effective solution. Understanding algorithms also helps students write programs that are faster, more organized, and efficient. Since IT involves software development, data processing, and technology solutions, having a strong understanding of algorithms is essential for becoming a better programmer and IT professional.

B. Convert each JavaScript code into a step-by-step algorithm using simple English or pseudocode.

1. for (let i = 1; i < 10; i *= 2)

Step 1: Start with i = 1.

Step 2: Check if i is less than 10.

Step 3: If it is, display the value of i.

Step 4: Multiply i by 2.

Step 5: Check again if i is less than 10.

Step 6: Repeat the process until i is no longer less than 10.

Step 7: Stop the loop.

Output: 1, 2, 4, 8

2. for (let i = 0; i < 10; i++)

Step 1: Start with i = 0.

Step 2: Check if i is less than 10.

Step 3: If it is, display the value of i.

Step 4: Increase i by 1.

Step 5: Check again if i is less than 10.

Step 6: Repeat the process until i reaches 10.

Step 7: Stop the loop.

Output: 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

C. Run both JavaScript codes. Based on the outputs and execution of the two codes, answer the following:

1. What did you observe about the sequence of numbers produced by the first code?

- The first code produces the numbers 1, 2, 4, and 8. The numbers increase by doubling each time, so the sequence grows quickly.

2. What did you observe about the sequence of numbers produced by the second code?

- The second code produces the numbers 0, 1, 2, 3,4 ,5 ,6,7, 8, 9. The numbers increase by 1 each time.

3. What is the main difference between the way the two loops progress?

- The main difference is how the value of i changes. The first loop multiplies i by 2 during every iteration, while the second loop increases i by 1.

4. How does the way an algorithm performs its steps affect its output?

- The way an algorithm performs its steps directly affects its output and the number of times the process is repeated. In the first loop, doubling the value makes the sequence grow faster and results in fewer numbers. In the second loop, increasing by 1 produces every number from 0 to 9. This shows that changing the steps or conditions of an algorithm can produce different results.